

ASSIGNMENT

Course: Full Stack Web Development

Course Code: 23CM4121

Assignment No.: 1

Student Name: J.Srikar

Roll No.: A24126552083

Date: 30-08-2026

TITLE

Implement Different Types of Inheritance in JavaScript

AIM

To understand and implement different types of inheritance in JavaScript using classes, the extends keyword, and mixins.

OBJECTIVES

- Implement single inheritance using JavaScript classes.
- Demonstrate multilevel inheritance through a chain of classes.
- Implement hierarchical inheritance with a common parent class.
- Demonstrate multiple inheritance behavior using mixins and Object.assign().
- Implement hybrid inheritance combining hierarchical and multilevel inheritance.
- Observe the output produced by each inheritance example.

CONCEPTS COVERED

- Classes and Objects
- extends keyword
- Prototype-based inheritance
- Single Inheritance
- Multilevel Inheritance
- Hierarchical Inheritance
- Multiple Inheritance using Mixins
- Hybrid Inheritance
- Object.assign()
- Method Reuse and Method Extension

TASK DESCRIPTION

Part A: Single Inheritance - Create a Bank base class and a Customer derived class. Demonstrate inherited bankName property and showBank() method along with child-specific showCustomer() method.

Part B: Multilevel Inheritance - Create Bank, Account, and SavingsAccount classes in a three-level inheritance chain and invoke methods across all levels.

Part C: Hierarchical Inheritance - Create a Loan base class and derive HomeLoan and CarLoan classes, each providing its own specialized loan details.

Part D: Multiple Inheritance - Define AccountDetails class and a Transaction mixin, combine them using Object.assign(), and demonstrate deposit functionality without multiple class inheritance.

Part E: Hybrid Inheritance - Combine hierarchical and multilevel inheritance structures using Employee, Manager, Cashier, and SeniorManager classes to verify complex multi-branch method resolution.

ALGORITHM / PROCEDURE

1. Define the Bank class with bankName property and a showBank() method.
2. Create Customer using extends Bank and add the showCustomer() method.
3. Create a Customer object and call both inherited and child methods.
4. Define Account extending Bank and SavingsAccount extending Account for multilevel inheritance.
5. Create a SavingsAccount object and invoke showBank(), showAccount(), and showBalance().
6. Define Loan as the parent class and derive HomeLoan and CarLoan as child classes.
7. Create HomeLoan and CarLoan objects and invoke their inherited and specialized methods.
8. Create AccountDetails class and define the Transaction mixin object with a deposit() method.
9. Use Object.assign(AccountDetails.prototype, Transaction) to merge mixin methods into AccountDetails.
10. Create an AccountDetails object and invoke showAccountDetails() and deposit().
11. Define Employee base class; derive Manager and Cashier (Hierarchical); then derive SeniorManager from Manager (Multilevel).
12. Create SeniorManager and Cashier objects to verify hybrid inheritance execution.

PROGRAM / SOURCE CODE

The following section contains the submitted JavaScript source code for the assignment:

```
// =====  
// TITLE: Implement Different Types of Inheritance in JavaScript  
// =====  
  
// 1. SINGLE INHERITANCE  
// One child class inherits from one parent class  
  
class Bank {  
    bankName = "ABC Bank";  
  
    showBank() {  
        console.log("Bank Name:", this.bankName);  
    }  
}  
  
class Customer extends Bank {  
    customerName = "Srikar";  
  
    showCustomer() {  
        console.log("Customer Name:", this.customerName);  
    }  
}  
  
console.log("\n--- SINGLE INHERITANCE ---");  
let customer = new Customer();  
customer.showBank();  
customer.showCustomer();  
  
// 2. MULTILEVEL INHERITANCE  
// Grandparent -> Parent -> Child  
  
class Account extends Bank {  
    accountNumber = "123456789";  
  
    showAccount() {  
        console.log("Account Number:", this.accountNumber);  
    }  
}  
  
class SavingsAccount extends Account {  
    balance = 50000;  
  
    showBalance() {  
        console.log("Balance:", this.balance);  
    }  
}  
  
console.log("\n--- MULTILEVEL INHERITANCE ---");  
let savings = new SavingsAccount();  
savings.showBank();  
savings.showAccount();  
savings.showBalance();  
  
// 3. HIERARCHICAL INHERITANCE  
// One parent -> Multiple child classes
```

```

class Loan {
    loanAmount = 100000;

    showLoan() {
        console.log("Loan Amount:", this.loanAmount);
    }
}

class HomeLoan extends Loan {
    showHomeLoan() {
        console.log("Home Loan");
    }
}

class CarLoan extends Loan {
    showCarLoan() {
        console.log("Car Loan");
    }
}

console.log("\n--- HIERARCHICAL INHERITANCE ---");

let home = new HomeLoan();
home.showLoan();
home.showHomeLoan();

let car = new CarLoan();
car.showLoan();
car.showCarLoan();

// 4. MULTIPLE INHERITANCE
// JavaScript does not support multiple inheritance using multiple classes directly.
// We achieve it using Mixins and Object.assign().

class AccountDetails {
    showAccountDetails() {
        console.log("Account Type: Savings Account");
    }
}

// Mixin
const Transaction = {
    deposit(amount) {
        console.log("Deposited:", amount);
    }
};

// Mixin assignment
Object.assign(AccountDetails.prototype, Transaction);

console.log("\n--- MULTIPLE INHERITANCE ---");

let account = new AccountDetails();
account.showAccountDetails();
account.deposit(10000);

// 5. HYBRID INHERITANCE
// Combination of two or more types of inheritance

```

```
class Employee {
  showEmployee() {
    console.log("Bank Employee");
  }
}

class Manager extends Employee {
  showManager() {
    console.log("Manager");
  }
}

class Cashier extends Employee {
  showCashier() {
    console.log("Cashier");
  }
}

class SeniorManager extends Manager {
  showSeniorManager() {
    console.log("Senior Manager");
  }
}

console.log("\n--- HYBRID INHERITANCE ---");
let sm = new SeniorManager();
sm.showEmployee();
sm.showManager();
sm.showSeniorManager();

let cashier = new Cashier();
cashier.showEmployee();
cashier.showCashier();
```

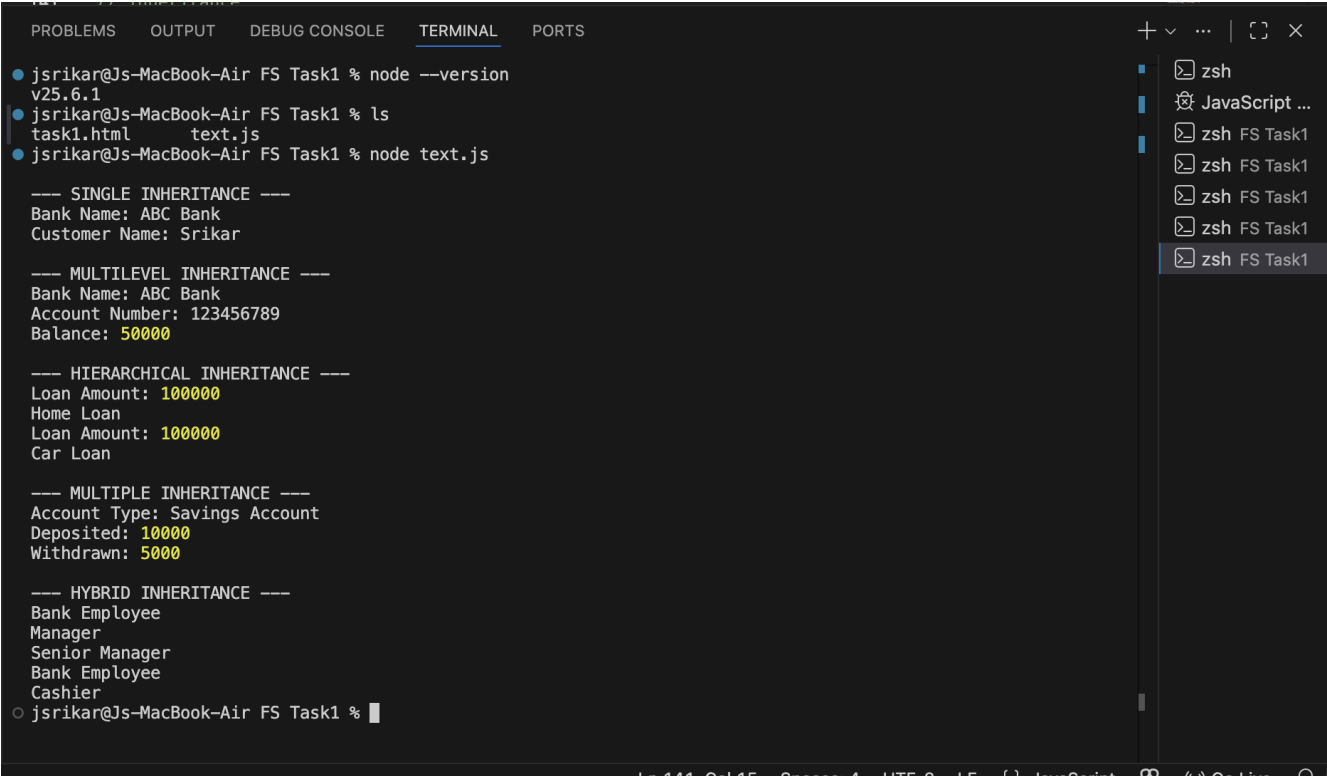
CODE EXPLANATION

JavaScript Element / Feature	Purpose
Bank / Customer	Customer extends Bank, demonstrating Single Inheritance where the child reuses bankName and showBank() while introducing customerName and showCustomer().
Bank / Account / SavingsAccount	SavingsAccount extends Account which extends Bank, illustrating Multilevel Inheritance. Instances can access properties and methods across all three hierarchy tiers.
Loan / HomeLoan / CarLoan	HomeLoan and CarLoan both inherit from Loan, demonstrating Hierarchical Inheritance with one parent and multiple specialized children sharing loanAmount.
AccountDetails / Transaction	Demonstrates Multiple Inheritance using Mixins. Since JS lacks multi-class extends, Object.assign() attaches deposit() from Transaction into AccountDetails.prototype.
Employee / Manager / Cashier / SeniorManager	Demonstrates Hybrid Inheritance by combining Hierarchical (Employee -> Manager, Cashier) and Multilevel (Employee -> Manager -> SeniorManager) patterns.
extends keyword	Standard ES6 keyword establishing prototype-based inheritance between parent and child classes.
Object.assign()	Copies enumerable properties and methods from source mixin objects directly onto the destination class prototype.
Class Field Initializers	Directly defines instance properties (bankName, balance, etc.) cleanly within the class body without explicit constructor boilerplates.

TEST CASES AND EXECUTION DATA

Test Case	Verification	Expected Result	Observed Result / Status
TC-01	Single inheritance	Bank Name and Customer Name messages are displayed.	Displayed - Pass
TC-02	Multilevel inheritance	Bank Name, Account Number, and Balance messages are displayed.	Displayed - Pass
TC-03	Hierarchical inheritance	Loan Amount and specialized Home/Car Loan messages are displayed.	Displayed - Pass
TC-04	Multiple inheritance	Account Type and Deposited amount messages are displayed.	Displayed - Pass
TC-05	Hybrid inheritance	Employee, Manager, and Senior Manager role hierarchy messages are displayed.	Displayed - Pass

OUTPUT



```
jsrikar@Js-MacBook-Air FS Task1 % node --version
v25.6.1
jsrikar@Js-MacBook-Air FS Task1 % ls
task1.html      text.js
jsrikar@Js-MacBook-Air FS Task1 % node text.js

--- SINGLE INHERITANCE ---
Bank Name: ABC Bank
Customer Name: Srikar

--- MULTILEVEL INHERITANCE ---
Bank Name: ABC Bank
Account Number: 123456789
Balance: 50000

--- HIERARCHICAL INHERITANCE ---
Loan Amount: 100000
Home Loan
Loan Amount: 100000
Car Loan

--- MULTIPLE INHERITANCE ---
Account Type: Savings Account
Deposited: 10000
Withdrawn: 5000

--- HYBRID INHERITANCE ---
Bank Employee
Manager
Senior Manager
Bank Employee
Cashier
jsrikar@Js-MacBook-Air FS Task1 %
```

RESULT

Thus, the different types of inheritance -- Single Inheritance, Multilevel Inheritance, Hierarchical Inheritance, Multiple Inheritance (using Mixins and `Object.assign()`), and Hybrid Inheritance -- were successfully implemented using JavaScript classes and mixins. The program demonstrates single inheritance, multilevel inheritance, hierarchical inheritance, multiple inheritance behavior using `Object.assign()`, and hybrid combinations. The supplied console output verifies the execution of all five examples.